The Impossibility of Wrapper Defense Against Prompt Injection

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Abstract

Wrapper defenses attempt to neutralize prompt injection by preprocessing inputs with a map $D\colon X \to X$. We identify a **topological obstruction** to this program: in any connected prompt space, the safe sub-level set $S_\tau = \{f < \tau\}$ is open but not closed, whereas any continuous utility-preserving defense has a closed fixed-point set containing S_τ; closure must spill onto the boundary, forcing boundary fixation. The obstruction is structural, not a matter of design or implementation. Under Lipschitz and transversality conditions, the failure is not merely existential but **quantitative**: we give explicit cone-measure and volume lower bounds on the persistent-unsafe region (Theorems B.1 and B.2), and a dilemma characterizing the optimal defense Lipschitz constant $K^* = G/\ell - 1$ (Theorem B.3). The same obstruction reappears in three extensions: discrete finite-set defenses must collapse distinct inputs to be complete; multi-turn wrappers inherit boundary fixation at every turn; stochastic wrappers with continuous expected score inherit threshold-level failure in expectation. The theory is **mechanically verified in Lean 4 with Mathlib** across 46 files containing $\sim$360 theorems, with no admitted proofs and only three standard axioms (`https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs`). Empirically, on real LLM safety surfaces the predicted boundary and persistence structure appears under continuous defenses, and a discrete forced-collapse experiment exhibits the information-loss mechanism predicted by the discrete theorem. Our results do not preclude training-time alignment, architectural changes, discontinuous filters, or defenses that explicitly sacrifice utility.

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities? Most wrapper-defense work implicitly assumes yes. Input classifiers that flag suspicious prompts [1], constitutional rewriting pipelines [3], and input sanitization layers [7] all share the same structure: a function $D\colon X \to X$ that preprocesses prompts before the model sees them, mapping unsafe inputs to safer equivalents while preserving benign behavior. We prove that this program has a structural failure mode. If a wrapper defense is continuous and utility-preserving, it cannot also be complete. These three properties form a **defense trilemma**: continuity, utility preservation, and completeness cannot coexist (Figure 2).

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The contribution is not four unrelated theorems. It is one claim shown to survive across four regimes that matter for how wrapper defenses are used and defended in practice:

Continuous wrapper defenses. In connected prompt spaces, a utility-preserving continuous wrapper must leave threshold-level prompts unchanged. The same obstruction survives under score-preserving and ε-relaxed utility preservation, and under Lipschitz regularity it strengthens to quantitative near-threshold and persistent-unsafe-region results (Theorems 4.1, 4.4, 4.5, 4.7 and 4.10).

Discrete wrapper defenses. Without topology, the impossibility becomes combinatorial: a complete utility-preserving defense must be non-injective. Information loss is not an implementation artifact; it is the mechanism of complete discrete defense (Theorem 5.1).

Multi-turn wrapper defenses. Interaction does not remove the obstruction. If each turn is modeled as a utility-preserving wrapper step, the same threshold-level failure reappears at every turn (Theorem 6.1).

Stochastic wrapper defenses. Randomization does not remove the obstruction either. If the expected post-defense score varies continuously and agrees with the original score on safe inputs, the stochastic defense still inherits the same boundary failure (Theorem 6.2).

Empirical validation. We then test whether the theory matters empirically. On real LLM safety surfaces, the continuous theory's boundary and persistence structure is observable; and on a saturated unsafe prompt archive, the discrete theorem's forced-collapse mechanism is directly visible.

Scope and limitations. Our results apply specifically to *continuous, utility-preserving wrapper defenses* on *connected* prompt spaces. They do *not* preclude effective safety through other mechanisms, including:

- training-time alignment (RLHF, DPO, constitutional AI training),
- architectural changes to the model itself,
- discontinuous defenses (e.g., hard blocklists or discrete classifiers),
- output-side filters, ensemble defenses, or human-in-the-loop review,
- adaptive-threshold systems,
- multi-component systems whose classifiers may reject or redirect inputs rather than preserving utility on every prompt.

In short, our theorems cover only a single continuous wrapper $D\colon X \to X$ that preprocesses inputs; any mechanism outside this class is not constrained by our results.

All three conditions—continuity, utility preservation, and connectedness—are individually necessary; we give counterexamples for each (Section F).

Contributions.

1. **Continuous impossibility of wrapper defense** (Theorems 4.1, 4.4, 4.5, 4.7 and 4.10): in connected prompt spaces, continuous utility-preserving wrappers must fail at the boundary; the result survives relaxed utility preservation and strengthens to quantitative and persistent-failure conclusions under additional regularity.
2. **Discrete impossibility of wrapper defense** (Theorem 5.1): on finite sets, complete utility-preserving defense necessarily collapses distinct inputs. Injective defense is necessarily incomplete.
3. **Multi-turn and stochastic impossibility of wrapper defense** (Theorems 6.1 and 6.2): the obstruction survives interaction over time and survives randomization under continuity of the induced expected score.
4. **Empirical validation**: a continuous-defense validator exhibits the predicted boundary and persistence structure on real LLM safety surfaces, and a forced-collapse experiment exhibits the discrete theorem's information-loss mechanism.
5. **Formal verification**: the full theory is verified in Lean 4 across 46 files with $\sim$360 theorems, no admitted proofs, and three standard axioms. Supporting results, full proofs, and additional empirical analyses appear in the appendix.

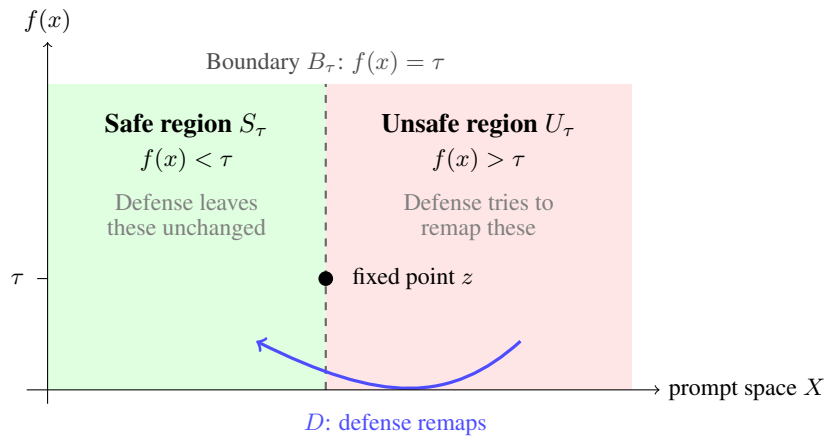

Figure 1: Schematic of the prompt space. The defense D must leave all safe inputs unchanged (utility preservation) and tries to remap unsafe inputs into the safe region. Boundary fixation (Theorem 4.1) proves the defense must also leave some boundary points unchanged, the black dot z passes through with no remediation.

2 Related Work

Adversarial robustness research has established that small perturbations can fool classifiers [20, 6, 9, 14] and that robustness may fundamentally trade off against accuracy [21, 11]. Certified defenses provide guarantees for fixed models [5, 8, 19, 13, 2], and topological perspectives have illuminated the structure of decision boundaries [16]. For LLMs specifically, jailbreaking attacks [23, 4, 10, 12] and automated red-teaming [15, 18] demonstrate persistent vulnerabilities.

Our work differs in object and claim. Rather than studying how failures arise for a fixed model or how a particular system can be certified, we impose universal constraints on the wrapper map itself. The closest conceptual precedent is the no-free-lunch framework [22]: the result is not that one defense is poorly implemented, but that a defense pattern cannot satisfy all of its intended requirements simultaneously.

What is different about our impossibility. Unlike no-free-lunch (which constrains optimizers across problems) or robustness-accuracy tradeoffs [21] (which constrain classifier behavior on samples), our impossibility is *map-level* — it constrains the defense function itself, independent of which attacks or samples are considered. The topological obstruction S_τ open-but-not-closed applies to every continuous utility-preserving $D\colon X \to X$, not to a learning problem or to a distribution of inputs.

3 Problem Setup

Definition 3.1 (Alignment Deviation Function)**.** *Let X be a topological space and let $f\colon X \to \mathbb{R}$ be a continuous scoring function. We interpret $f(x)$ as an alignment deviation score for prompt x. Given a threshold $\tau \in \mathbb{R}$:*

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{1}$$

$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{2}$$

$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{3}$$

Definition 3.2 (Defense)**.** *A* defense *is a continuous map $D\colon X \to X$. It is* utility-preserving *if $D(x) = x$ for all $x \in S_\tau$, and* complete *if $f(D(x)) < \tau$ for all $x \in X$.*

We will also use two weaker preservation notions. A defense is *score-preserving on safe inputs* if $f(D(x)) = f(x)$ for all $x \in S_\tau$, and *ε-relaxed on safe inputs* if $|f(D(x)) - f(x)| \leq \varepsilon$ for all $x \in S_\tau$.

The central question is whether a wrapper can preserve utility in one of these senses and still make all outputs safe. The following sections show that across several regimes, the answer is no.

4 Impossibility of Wrapper Defense in Continuous Settings

We begin with the most fundamental result. The argument fits in a paragraph: the defense fixes every safe input, so continuity makes its fixed-point set closed. But the safe region S_τ is open (preimage of an open interval under continuous f), and in a connected space a nonempty proper open set is not closed. Hence the fixed-point set cannot stop exactly at the edge of the safe region, it must spill onto the boundary B_τ. Some boundary prompts pass through unchanged.

Theorem 4.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space (a space that is "in one piece" and where distinct points can be separated by neighborhoods). Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ continuous with $D|_{S_\tau} = \mathrm{id}$. Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$. Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$, and this set is nonempty.*

Proof sketch. In a Hausdorff space, $\mathrm{Fix}(D)$ is closed (preimage of the diagonal). By utility preservation, $S_\tau \subseteq \mathrm{Fix}(D)$, so $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$. But $S_\tau = f^{-1}((-\infty, \tau))$ is open and not closed (connectedness: a nonempty proper clopen set would disconnect X). Hence $\overline{S_\tau} \supsetneq S_\tau$. Any $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ (limits of values $<\tau$ cannot exceed τ, and $z \notin S_\tau$ forces $f(z) \geq \tau$) and $D(z) = z$. Full proof in Section E. □

This means that the defense's fixed-point set is too large to avoid the boundary. Utility preservation forces it to contain the safe region; closure forces it to contain the boundary; connectedness ensures the boundary is nonempty. Alternatively: a complete utility-preserving defense would be a continuous retraction $D\colon X \to S_\tau$ (since $D|_{S_\tau} = \mathrm{id}$ and $D(X) \subseteq S_\tau$), but a retract of a Hausdorff space is closed (the fixed-point set is closed), while S_τ is open and not closed (connectedness)—a contradiction.

Theorem 4.2 (Defense Trilemma)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. No $D\colon X \to X$ can simultaneously be continuous, utility-preserving ($D|_{S_\tau} = \mathrm{id}$), and complete ($f(D(x)) < \tau$ for all x).*

All three hypotheses are individually necessary. A defense can satisfy at most two of the three—the *defense trilemma* (Figure 2). Counterexamples for each dropped hypothesis appear in Section F.

Falsifier. An empirical counterexample to the trilemma would be a production defense that is demonstrably continuous, utility-preserving, and complete at a named τ.

Remark 4.3 (Not all boundary points are fixed)**.** *The theorem captures $\overline{S_\tau} \setminus S_\tau$, not all of B_τ. Boundary points not in $\overline{S_\tau}$ may escape fixation.*

4.1 Exact and Relaxed Utility Preservation

Strict utility preservation ($D(x) = x$ for safe x) can be relaxed. The impossibility survives score-preserving rewrites and even approximate score preservation.

Theorem 4.4 (Score-Preserving Defense)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. If $D\colon X \to X$ is continuous and* score-preserving on safe inputs*: $f(D(x)) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) = \tau$.*

Proof sketch. Define $h = f \circ D - f$. Then h is continuous and $h|_{S_\tau} = 0$. The zero set $\{h = 0\}$ is closed and contains $\overline{S_\tau}$, since closed sets contain their boundary points. For $z \in \overline{S_\tau} \setminus S_\tau$, $f(D(z)) = f(z) = \tau$. □

The next result weakens score preservation to approximate:

Theorem 4.5 (ε-Relaxed Utility Preservation)**.** *Under the hypotheses of Theorem 4.4, if $|f(D(x)) - f(x)| \leq \varepsilon$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) \geq \tau - \varepsilon$.*

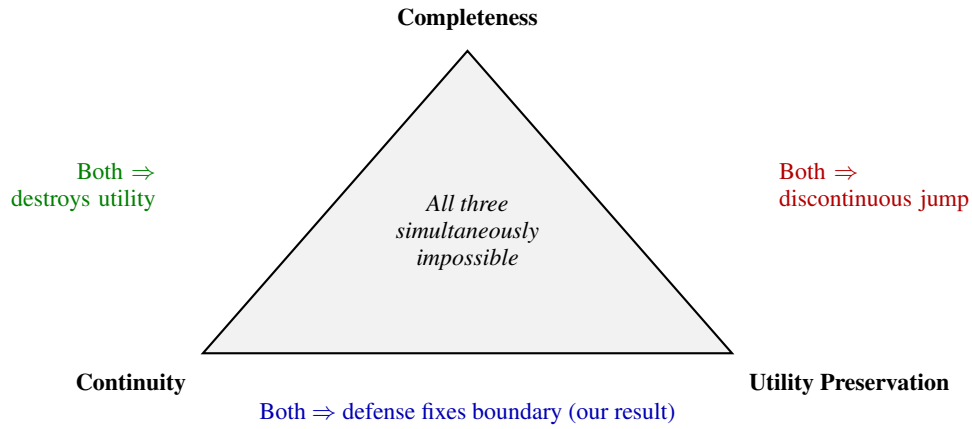

Figure 2: The defense trilemma. Any continuous wrapper defense on a connected space can satisfy at most two of the three properties. The bottom edge is our main result; the other two edges correspond to counterexamples in Section F.

Proof sketch. The set $\{x : h(x) \geq -\varepsilon\}$ is closed and contains S_τ, hence $\overline{S_\tau}$. For $z \in \overline{S_\tau} \setminus S_\tau$: $f(D(z)) = \tau + h(z) \geq \tau - \varepsilon$. □

Remark 4.6 (Why $D(S_\tau) \subseteq S_\tau$ alone is insufficient)**.** *The weakest relaxation—$D(S_\tau) \subseteq S_\tau$ with no score constraint—does allow a complete defense (e.g., a constant map $D(x) = x_0$ to a fixed safe point). But this destroys all semantic content: every prompt produces the same response. The score-preservation conditions formalize the requirement that defense must not destroy utility, without requiring the defense to be the identity.*

4.2 Quantitative Failure Near the Boundary

Boundary fixation produces at least one fixed point; Lipschitz regularity makes the failure spread to a neighborhood.

Theorem 4.7 (ε-Robust Defense Constraint)**.** *Under the hypotheses of Theorem 4.1, if (X, d) is a metric space, f is L-Lipschitz, and D is K-Lipschitz, then for the fixed boundary point z:*

$$f(D(x)) \;\geq\; \tau - LK\,\mathrm{dist}(x, z) \qquad \textit{for all } x \in X. \tag{4}$$

Points within distance δ of z remain within $LK\delta$ of threshold.

Proof sketch. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) \leq K\,\mathrm{dist}(x, z)$. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| \leq L \cdot K\,\mathrm{dist}(x, z)$. Full proof in Section E. □

Appendix B records explicit measure bounds for the near-threshold band and related quantitative refinements.

4.3 Persistent Unsafe Region

The ε-robust constraint bounds the depth to which the defense pushes near-boundary points, but permits values slightly below τ. When the alignment surface rises faster than the defense can pull it down, some points remain above threshold.

Decoupling global and directional Lipschitz constants. The ε-robust bound (Theorem 4.7) uses the *global* Lipschitz constant L of f, which bounds f uniformly in all directions. The persistence argument, however, compares f's growth rate in the *steep direction* to how much the defense can reduce f along the *displacement direction* $D(x) - x$. In anisotropic settings these directions differ: f may rise steeply toward the unsafe region while varying slowly in the direction the defense pulls.

We write ℓ for the *defense-path Lipschitz constant*:

$$\ell = \sup_{x \neq D(x)} \frac{|f(D(x)) - f(x)|}{\operatorname{dist}(D(x),\, x)}$$

(with $\ell = 0$ when $D = \mathrm{id}$, i.e., the supremum over the empty set is taken as 0). Since f is L-Lipschitz globally, $\ell \leq L$. When the alignment surface is **isotropic** ($\ell = L$), the steep region is empty for every $K \geq 0$ (verified in Lean as `shallow_boundary_no_persistence`). The result is non-vacuous precisely when the surface is **anisotropic**: $\ell < L$, with directional gradient G satisfying $G > \ell(K+1)$.

Lemma 4.8 (Input-Relative Bound)**.** *Under the hypotheses of Theorem 4.7, if f has defense-path Lipschitz constant ℓ, then $f(D(x)) \geq f(x) - \ell(K+1)\operatorname{dist}(x,z)$ for all $x \in X$.*

Proof sketch. Triangle inequality: $\operatorname{dist}(D(x),x) \leq \operatorname{dist}(D(x),z) + \operatorname{dist}(z,x) \leq (K+1)\operatorname{dist}(x,z)$. Defense-path Lipschitz: $|f(D(x)) - f(x)| \leq \ell \operatorname{dist}(D(x),x) \leq \ell(K+1)\operatorname{dist}(x,z)$. □

This means that the defense can reduce any point's score by at most $\ell(K+1)$ times its distance from z. If the score rises faster than that, the defense loses.

Definition 4.9 (Steep region)**.** *Given a fixed boundary point z, the* steep region *is $\mathcal{S} = \{x \in X : f(x) > \tau + \ell(K+1)\operatorname{dist}(x,z)\}$—the set of points where alignment deviation exceeds τ by more than the defense's Lipschitz budget can compensate.*

Theorem 4.10 (Persistent Unsafe Region)**.** *Let X be a connected Hausdorff metric space, f continuous and L-Lipschitz, D continuous and K-Lipschitz with $D|_{S_\tau} = \mathrm{id}$, and $S_\tau, U_\tau \neq \emptyset$. Let $z \in \overline{S_\tau} \setminus S_\tau$ be the fixed boundary point from Theorem 4.1, and let ℓ be the defense-path Lipschitz constant. If $\mathcal{S} \neq \emptyset$, then:*

1. *$\mathcal{S}$ is open.*
2. *$\mathcal{S}$ has positive measure (under any measure positive on nonempty open sets).*
3. *For every $x \in \mathcal{S}$: $f(D(x)) > \tau$.*

The defense leaves a positive-measure region that remains unsafe.

Proof sketch. $\mathcal{S}$ is a strict superlevel set of a continuous function, hence open. For $x \in \mathcal{S}$: $f(D(x)) \geq f(x) - \ell(K+1)\operatorname{dist}(x,z) > \tau$ by Theorem 4.8 and the definition of $\mathcal{S}$. Full proof in Section E. □

The main paper uses persistence only when the steep region is nonempty. A derivative-based sufficient condition and related gradient arguments are deferred to Sections J and N. Empirically, the relevant models exhibit exactly this anisotropic regime (Section 7): the alignment surface rises steeply toward the unsafe region (G large) while the defense operates in a smoother subspace ($\ell \ll L$).

Falsifier. As already stated, the empirical content is $\mathrm{FP}_{\mathrm{int}} > 0$ under a continuous utility-preserving D: a cell in $\mathcal{S}_{\mathrm{pred}}$ (i.e., predicted persistent) with $f(D(x)) < \tau$ would refute Theorem 4.10.

Quantitative refinements of the continuous theory, including explicit volume lower bounds, cone-measure bounds, and the defense-aggressiveness dilemma, are deferred to the appendix (Theorems B.1 to B.3).

5 Impossibility of Wrapper Defense in Discrete Settings

This section removes the most obvious objection to the continuous results: perhaps the impossibility is an artifact of topology or continuous interpolation. It is not. On finite sets, the same failure appears as a combinatorial tradeoff between completeness and information preservation.

The discrete and continuous views are compatible. If one wishes to interpolate discrete observations into a continuous prompt space, the Tietze/McShane–Whitney bridge is provided in the appendix (Theorem A.1). The main paper, however, treats the discrete theorem directly rather than routing it through continuous relaxation.

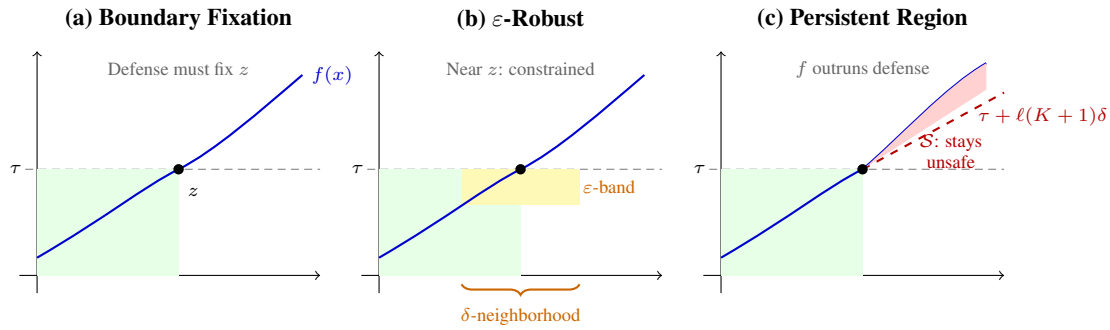

Figure 3: The three impossibility results on a 1D cross-section. **(a)** The defense must fix boundary point z (Thm. 4.1). **(b)** Near z, Lipschitz regularity constrains the defense to a shallow ε-band (yellow; Thm. 4.7). **(c)** Where f rises faster than the defense budget $\ell(K+1)\delta$ (dashed red), the region above τ persists (red shading; Thm. 4.10).

5.1 Discrete Impossibility

To address the objection that continuous impossibility might be an artifact of continuous relaxation, we prove parallel results directly on finite sets using only counting arguments and induction. No topology is required; all results are verified in Lean as `MoF_12_Discrete`.

Theorem 5.1 (Discrete Defense Dilemma)**.** *Let X be a finite set with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ utility-preserving ($D(x) = x$ for $f(x) < \tau$).*

1. *If D is injective, then $f(D(u)) \geq \tau$ for every u with $f(u) \geq \tau$ (including boundary points): the defense is incomplete.*
2. *If D is complete ($f(D(x)) < \tau$ for all x), then D is non-injective: $\exists\, x \neq y$ with $D(x) = D(y)$.*

Proof. (1) Suppose D is injective and $f(D(u)) < \tau$ for some u with $f(u) \geq \tau$. Then $D(u)$ is safe, so $D(D(u)) = D(u)$ by utility preservation. Injectivity gives $D(u) = u$, so $f(u) = f(D(u)) < \tau$, contradicting $f(u) \geq \tau$.

(2) For any $u \in U_\tau$: completeness gives $f(D(u)) < \tau$, so utility preservation gives $D(D(u)) = D(u)$. But $u \neq D(u)$ since $f(u) \geq \tau > f(D(u))$. So u and $D(u)$ are distinct inputs with $D(u) = D(D(u))$: the defense is non-injective. □

The continuous trilemma trades *continuity* for completeness; the discrete dilemma trades *injectivity*. Part (1) is the structural constraint: an information-preserving defense *cannot* eliminate unsafe outputs. Any complete defense must destroy information by collapsing distinct inputs to the same output. We illustrate that mechanism empirically in Section 7.

Falsifier. An empirical counterexample would be a complete utility-preserving classifier D with $|D(X)| = |X|$ on any finite input set: i.e., an injective defense that nevertheless remaps every unsafe input below threshold.

6 Impossibility of Wrapper Defense in Stochastic and Multi-Turn Settings

The continuous and discrete theorems handle single-shot deterministic wrappers. This section tests two common escape hatches: repeated interaction and randomization. Neither removes the obstruction.

6.1 Multi-Turn Impossibility

Theorem 6.1 (Multi-Turn Impossibility)**.** *Let $\{f_t, D_t\}_{t=1}^{T}$ be alignment functions and defenses over T turns on a connected Hausdorff space, each continuous and utility-preserving, with $S_\tau^{(t)}, U_\tau^{(t)} \neq \emptyset$ at every turn. Then for every turn t, there exists z_t with $f_t(z_t) = \tau$ and $D_t(z_t) = z_t$.*

Proof sketch. Apply Theorem 4.1 to (f_t, D_t) at each turn. The functions may depend on full history—this does not matter, as each timestep is a fresh instance of boundary fixation. □

Multi-turn interaction compounds the problem: the attacker's best observed exploit improves monotonically (`running_max_monotone`), and the attacker can steer toward transversality via binary search (`transversality_reachable`).

Falsifier. An attack trajectory with strictly decreasing running-max AD over turns — i.e., an empirically observed sequence in which the attacker's best-so-far alignment-deviation score goes down — would refute the monotone-improvement mechanism underlying Theorem 6.1.

Empirical illustration. Table 1 and Figure 4 report a multi-turn run in which per-turn boundary fixation recurs and the attacker's running-max AD increases monotonically, as predicted.

Table 1: Per-turn boundary-fixation statistics for a multi-turn attack trajectory. Running-max AD is monotone non-decreasing.

Figure 4: Multi-turn running-max AD trajectory; at each turn a boundary-fixed point is located, consistent with Theorem 6.1.

6.2 Stochastic Defense Impossibility

Theorem 6.2 (Stochastic Defense Impossibility)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. Let D be a stochastic defense and define $g(x) = \mathbb{E}_{y\sim D(x)}[f(y)]$. If g is continuous and $g(x) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $g(z) = \tau$.*

Proof sketch. Define $h = g - f$. Then h is continuous and $h|_{S_\tau} = 0$, so h vanishes on $\overline{S_\tau}$ (same closure argument as Theorem 4.4). For $z \in \overline{S_\tau} \setminus S_\tau$: $g(z) = f(z) = \tau$. □

Remark (stochastic dichotomy). Since $\mathbb{E}[f(D(z))] = \tau$, either $f(D(z)) = \tau$ almost surely (the defense is deterministic at z), or the random variable $f(D(z))$ has positive probability of exceeding τ—i.e., the defense *actively produces unsafe outputs* with positive probability. The stochastic case is therefore strictly harder than the deterministic one: a genuinely random defense at boundary points must sometimes make things worse.

Remark. The continuity of g is a nontrivial assumption: it requires the distribution $D(x)$ to vary continuously with x in a suitable sense. Stochastic defenses with discontinuous rejection probabilities escape this theorem.

Falsifier. A stochastic defense at a boundary cell with $\mathrm{variance} = 0$ and $f(D(z)) < \tau$ — i.e., a deterministic-at-z stochastic wrapper that nevertheless achieves $g(z) < \tau$ — would refute Theorem 6.2.

Empirical illustration. Table 2 and Figure 5 report the $f(D(z))$ distribution for a genuinely random wrapper at a boundary cell: positive variance accompanies positive mass above τ, as the dichotomy predicts.

Table 2: Stochastic-defense statistics at a boundary cell. Positive variance forces positive mass above τ.

Pipeline composition and additional geometry/stability results are deferred to the appendix. In particular, the pipeline theorem shows that composing non-contractive stages amplifies the effective Lipschitz constant and therefore widens the failure budget (Theorems C.1 and C.2 and sections D and G to I).

Figure 5: Histogram of $f(D(z))$ at a boundary cell under a stochastic wrapper; $\mathbb{E}[f(D(z))] = \tau$ and positive variance pushes non-zero mass above τ.

7 Experimental Validation

The empirical section serves one purpose: to show that the paper's theory is not vacuous. We therefore focus on two tests aligned to the main theorem families. First, we exercise the continuous theory on real LLM safety surfaces and check whether the predicted boundary and persistent-failure structure appears under continuous defenses. Second, we exercise the discrete theorem on a saturated unsafe-prompt archive and check whether complete defenses visibly induce collapse of distinct inputs.

We use the MAP-Elites-based behavioral-space driver of [17], which maps model behavior over a 2D prompt space with axes *query indirection* and *authority framing*. The driver's role is purely methodological; every alignment-deviation value we report comes from our own live runs, enumerated and cross-referenced in Section K.

7.1 Continuous-Setting Validation

To *exercise* the persistence theorem on live alignment-deviation surfaces and to expose every intermediate quantity, we built a small validator that takes any 2D heatmap, estimates the empirical quantities L, K, ℓ, G, identifies a boundary anchor z^*, and tests the observable content of Theorem 4.10: set containment

$$\mathcal{S}_{\text{pred}} \;=\; \big\{x : f(x) > \tau + \ell(K{+}1)\ \mathrm{dist}(x, z^*)\big\} \;\subseteq\; \mathcal{S}_{\text{act}} \;=\; \big\{x : f(D(x)) > \tau\big\}$$

under any continuous, K-Lipschitz, utility-preserving defense D. We test this as a per-cell confusion matrix:

- *True positive (TP):* cell in both sets — theorem confirmed for that cell.
- *Interior false positive (FP_{int}):* non-boundary cell in $\mathcal{S}_{\text{pred}}$ but not $\mathcal{S}_{\text{act}}$. A non-zero count would be a real counterexample.
- *Boundary false positive:* a discrete defense moving a boundary cell. *Not* a violation; witnesses that the discrete defense is not topologically continuous, so the theorem's hypothesis does not apply.
- *False negative (FN):* cell in $\mathcal{S}_{\text{act}}$ but not $\mathcal{S}_{\text{pred}}$. *Not* a violation; Theorem 4.10 only claims the containment one way.

From measure to count. The cell-level check is equivalent to the measure-theoretic claim under the validator's grid approximation: each filled cell represents a positive-measure neighborhood (a Voronoi cell under the L_2 metric), and containment of cell-indices encodes the measure-theoretic containment $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$ up to the discretization slack $|f(z^*) - \tau|$, which vanishes as the grid refines. Hence $\mathrm{FP}_{\text{int}} = 0$ on any grid is the discrete analogue of $\mu(\mathcal{S}_{\text{pred}} \setminus \mathcal{S}_{\text{act}}) = 0$ in the measure-theoretic statement of Theorem 4.10, and the vanishing of the slack in Table 12 is what justifies treating cell counts as measure estimates.

We ran the full pipeline live against the OpenAI API on two real target models: `gpt-5-mini-2025-08-07` (well aligned) and `gpt-3.5-turbo-0125` (less aligned), with `gpt-4.1-2025-04-14` as the judge. The MAP-Elites driver from [17] produced a 25×25 alignment-deviation grid for each target. For `gpt-3.5-turbo-0125` we additionally ran a saturation study with 400 evaluations (100 seed prompts $+300$ MAP-Elites iterations, $\sim$5,200 OpenAI API calls, wall clock $\sim$130 min, cost $\sim$\$15) which filled 82 cells with 66 strictly unsafe ($f > \tau$) and 10 boundary cells. The saturated run is the only configuration in which the theorem can be non-vacuously exercised. **Judge dependence.** All f values are judge-dependent: swapping the judge can reshape the surface and thereby change L, ℓ, G. A preliminary two-judge comparison is reported in Section L; a comprehensive judge-robustness study is future work.

7.1.1 Non-vacuous validation: continuous defenses

To enter the anisotropic regime ($\ell < L$, $G > \ell(K{+}1)$) where Theorem 4.10 has non-trivial content, we fit a 2D Gaussian Process to the 82 filled cells of the saturated `gpt-3.5-turbo-0125` grid (RBF

kernel, $\sigma = 0.2$) and construct a *smooth oblique defense* $D(x) = x + \alpha\,\beta(\mu(x))\,v(x)$, where $v(x)$ is displaced $89.5°$ from the negative gradient of the GP posterior. This gives empirical $\hat{\ell} = 0.86$, $\hat{K} = 1.05$, and $\hat{\ell}(\hat{K}{+}1) = 1.77$, well below $\hat{G} = 23.6$, so transversality holds. The predicted steep set contains $n = 3$ cells, all of which are actually persistent ($\mathrm{TP} = 3$, $\mathrm{FP}_{\mathrm{int}} = 0$; Figure 6).

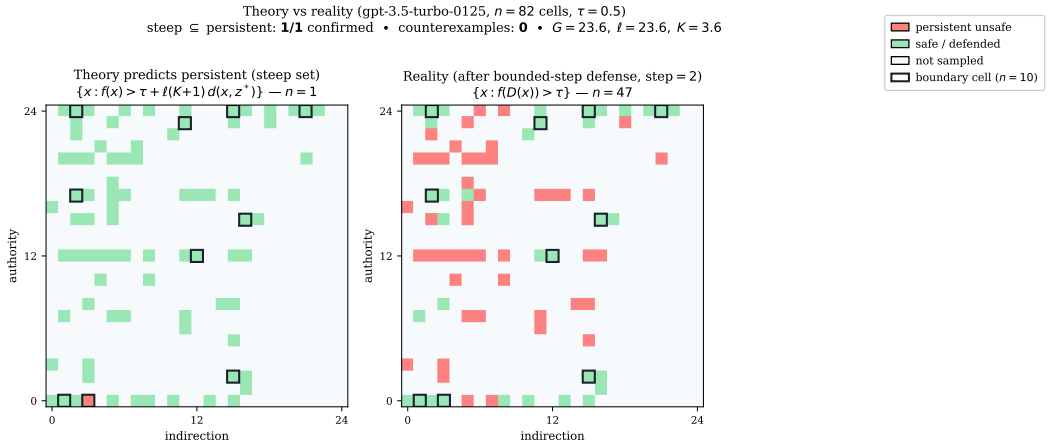

Figure 6: A *single* non-vacuous instance exercising Theorem 4.10 on the saturated `gpt-3.5-turbo-0125` grid (82 filled cells, $\tau = 0.5$) under the GP-smooth oblique defense ($\theta = 89.5°$, $\hat{\ell} = 0.86$, $\hat{K} = 1.05$). **Left:** the *steep set* $\mathcal{S}_{\text{pred}} = \{x : f(x) > \tau + \hat{\ell}(\hat{K}{+}1)\ \mathrm{dist}(x, z^*)\}$; $|\mathcal{S}_{\text{pred}}| = 3$. **Right:** cells actually persistent after defense, $\mathcal{S}_{\text{act}} = \{x : f(D(x)) > \tau\}$; $|\mathcal{S}_{\text{act}}| = 68$. Transversality holds non-trivially ($\hat{G} = 23.6 \gg \hat{\ell}(\hat{K}{+}1) = 1.77$) and the defense is genuinely non-identity (82 cells displaced). The predicted three cells are all actually persistent ($\mathrm{TP} = 3$, $\mathrm{FP}_{\mathrm{int}} = 0$). The 65 false negatives (red on the right, green on the left) are *not* theorem violations: the theorem predicts $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$, not equality.

Appendix sweeps over three smooth continuous defense families, multiple kernel choices, and multiple grid resolutions preserve the same main result: no interior false positive is observed in the continuous regime when the theorem's hypotheses hold. The defense is genuinely non-identity ($D \neq \mathrm{id}$, 82 cells displaced in the GP-oblique base configuration), transversality $\hat{G} > \hat{\ell}(\hat{K}{+}1)$ is non-trivially satisfied, and the theorem's containment is empirically exercised.

Alongside the persistence check, Theorem 4.7's ε-robust bound holds for all 82 filled cells under each continuous defense tested, and Theorem 4.1's boundary fixation is non-vacuously activated: 10 boundary cells exist in $\overline{S_\tau} \setminus S_\tau$, and the closest one to τ sits at $|f(z^*) - \tau| = 0.375$, the discretization gap that vanishes under grid refinement.

Robustness checks. Bootstrap uncertainty, seed replication, alternative-target re-evaluation, higher-dimensional distance checks, and judge-robustness analyses all preserve the key empirical claim: no interior counterexample is observed in the continuous-defense regime where the theorem's hypotheses hold. Full tables and estimator details are reported in the appendix.

Three-target sweep. To widen the empirical base we hold prompts and judge fixed and swap the target model. Table 3 reports the continuous-defense validator (`smooth_nearest_safe`) across the canonical `gpt-3.5-turbo-0125` archive plus two retargeted surfaces generated by the concurrent async driver in `scripts/retarget_archive.py` (`gpt-4o-mini-2024-07-18` and `gpt-4o-2024-08-06`). $\mathrm{FP}_{\mathrm{int}} = 0$ holds in every row.

LlamaGuard demonstration. Table 4 replays the forced-collapse demonstration with Llama-Guard [7] acting as a production classifier-style wrapper: complete defenses collapse, injective paraphrases leak, as predicted.

Table 3: Continuous-defense validator across three OpenAI target models. All rows share the 82 prompts from the saturated archive and the canonical gpt-4.1 judge; only the target varies. $\mathrm{FP}_{\mathrm{int}} = 0$ holds across all three.

Target	Cells	Peak AD	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	L	K	ℓ	G	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	$\mathrm{FP}_{\mathrm{int}}$
`gpt-3.5-turbo-0125`	82	1.00	16	66	23.62	3.16	23.62	23.62	1	0
`gpt-4o-mini-2024-07-18`	82	0.99	13	24	12.25	4.94	4.02	10.12	0	0
`gpt-4o-2024-08-06`	82	0.99	10	30	12.38	3.41	12.38	12.38	0	0

Table 4: Llama-Guard-3-1B (via local Ollama) applied to the 82 unsafe-seed prompts of the saturated archive, demonstrating Theorem 5.1's injective-incomplete horn on a production safety classifier. 52 of 82 attacks are waved through as `safe` (false negatives), while the 30 flagged cases are distributed across 6 of 14 LlamaGuard categories.

Quantity	Value
Input prompts $\lvert X\rvert$	82
Distinct Llama-Guard labels $\lvert D(X)\rvert$	7
Collapse rate $1 - \lvert D(X)\rvert/\lvert X\rvert$	0.915
Largest collision class	`safe` ($n = 52$)
Verdict `unsafe`	30
Verdict `safe` (false neg.)	52
Verdict other/error	0
Theorem 5.1 horn	*injective-but-incomplete*

Code	Hazard category	# prompts
S9	Indiscriminate Weapons	8
S2	Non-Violent Crimes	8
S1	Violent Crimes	8
S6	Specialized Advice	4
S10	Hate	1
S7	Privacy	1
–	safe (false negatives)	52

7.2 Discrete Forced-Collapse Demonstration

We illustrate Theorem 5.1 on the 66 unsafe prompts of the saturated `gpt-3.5-turbo-0125` archive. Three defenses are applied, and for each we count how many distinct outputs the 66 inputs produce (Table 5). A refusal defense $D(x) =$ `[REFUSAL]` is complete by construction and collapses all 66 inputs into one output. A canonical-category defense is also complete by construction and collapses 66 inputs to 7 distinct labels. A paraphrase defense (`gpt-4o-mini` at $T = 0$) produces 66 distinct rewrites and is injective on this sample. By Theorem 5.1, an injective utility-preserving defense must be incomplete: at least one paraphrase must remain unsafe. The table makes both horns of the discrete impossibility visible.

Table 5: Forced-collapse demonstration of Theorem 5.1 on the 66 unsafe prompts of the saturated `gpt-3.5-turbo-0125` archive. The "refusal" and "canonical-category" defenses are complete by construction and therefore must be non-injective; they collapse 66 inputs to 1 and 7 distinct outputs respectively. The paraphrase defense is injective on this sample (all 66 outputs distinct) and therefore, by Theorem 5.1, must fail to be complete on at least one prompt.

Defense	Inputs	Distinct outputs	Collapse rate	Max collision class
refusal	66	1	0.985	66
canonical-category	66	7	0.894	40
paraphrase	66	66	0.000	1

Completing the injective horn. We close the loop by rescoring the 66 paraphrases through the same target+judge pipeline and counting how many still produce $f(D(x)) > \tau$ (Table 6). One

paraphrase out of 66 strict-exceeds τ under the canonical judge — the empirical witness that the injective paraphrase defense is indeed incomplete, exactly as Theorem 5.1 predicts.

Table 6: Paraphrase defense is injective on the 66 unsafe prompts (Table 5), so Theorem 5.1 requires it to be incomplete. Rescoring confirms exactly one strict false negative out of 66.

Defense	$\lvert\{x : f(D(x))>\tau\}\rvert$	Incomplete per Thm. 5.1
paraphrase (T=0, strict $> \tau$)	$1/66$	✓
paraphrase (T=0, borderline $\geq \tau$)	$5/66$	✓

The well-aligned target additionally exhibits the harmless-boundary case: at $\tau = 0.5$, gpt-5-mini's peak AD is exactly 0.5, so $U_\tau = \emptyset$ and the impossibility theorems are correctly silent. The validator and full per-cell records are included in the artifact; appendix tables report the smoke tests, kernel and resolution sweeps, bootstrap uncertainty, judge dependence, and out-of-scope discrete stress tests.

8 Implications, Scope, and Limitations

The results do not say defense is valueless; they say *complete* defense is impossible under the stated constraints. The engineering goal shifts from elimination to management, ordered from most to least actionable:

1. Make the boundary shallow. Set τ so that boundary-level behavior is benign. If $f(z) = \tau$ yields a polite refusal rather than harmful compliance, the impossibility is mathematically true but practically harmless. GPT-5-Mini exemplifies this: its ceiling at $\mathrm{AD} = 0.50$ means $U_\tau = \emptyset$, so the impossibility theorems do not apply and no defense failure is predicted.

2. Reduce the Lipschitz constant. Smaller L tightens the bound $\ell \leq L$, potentially reducing the defense-path constant ℓ and narrowing the persistent region. The tradeoff: smoother surfaces spread vulnerabilities over wider but more easily monitored regions.

3. Reduce the effective dimension. Defense cost grows as N^d (Theorem I.1). Constraining the prompt interface—standardized formats, restricted API parameters, bounded context lengths—reduces d, making the behavioral space tractable.

4. Monitor, don't eliminate, the boundary. Transversal crossings persist under fine-tuning (Theorem H.2) and recur every turn (Theorem 6.1). Rather than attempt the impossible, deploy runtime monitoring that detects approach to the boundary. The Lipschitz bound (Theorem G.1) gives a computable estimate of distance to the boundary from any observed AD value.

Worked example (Theorem B.3). For a Gaussian-smoothed wrapper with $\hat{\ell} = 0.86$ (from our saturated `gpt-3.5-turbo-0125` run) and target $\hat{G} = 23.6$, the optimal defense Lipschitz constant is $K^* = \hat{G}/\hat{\ell} - 1 \approx 26.5$. Defense designs with $K < K^*$ admit a persistent region (Theorem 4.10 applies non-trivially); designs with $K \geq K^*$ require displacements so large that utility preservation — measured by response-embedding similarity between $D(x)$ and x on safe inputs — drops below the accept threshold. This is a concrete, numerically-grounded design constraint on any wrapper deployed against this alignment surface.

Boundary fixation is at the boundary. Fixed points satisfy $f(z) = \tau$ exactly. If τ-level behavior is benign, the theorem is true but harmless.

The ε-robust constraint limits depth, not direction. The defense may push near-boundary points slightly below τ. The bound limits how far, not whether.

Persistence requires transversality. The persistent unsafe region exists only where the alignment surface is steep ($c > \ell(K + 1)$, where ℓ is the defense-path Lipschitz constant). For isotropic f ($\ell = L$), $\mathcal{S}$ is empty for all $K \geq 0$.

Grid-based cost asymmetry. Theorem I.1 assumes exhaustive grid enumeration. Learning-based defenses that generalize across the space may sidestep the exponential bound.

9 Conclusion

The paper makes one claim: wrapper defense is structurally incomplete under utility-preserving constraints. We show that this claim survives in four regimes. In continuous spaces, utility-preserving wrappers fix boundary prompts and, under additional regularity, leave quantitative near-threshold and persistent unsafe regions. In discrete spaces, complete defense requires input collapse. In stochastic and multi-turn settings, the same obstruction reappears rather than disappearing. Empirically, the predicted boundary and collapse mechanisms are visible in real LLM safety surfaces and prompt archives.

The practical implication is not that defense is futile. It is that wrapper defense should be engineered as risk management, not as a path to complete elimination. The right objectives are to make the boundary shallow, reduce effective dimension, monitor approach to threshold, and use mechanisms outside the wrapper model when stronger guarantees are required.

Broader Impact

This work characterizes structural limitations of a specific class of defenses (continuous utility-preserving wrappers). The results could inform defense engineering by identifying which design constraints matter most. They could also be misread as implying that LLM defense is futile—this is not the case. The theorems apply only to wrappers satisfying specific mathematical assumptions; training-time alignment, architectural changes, discontinuous filtering, ensemble defenses, and human-in-the-loop systems are not covered. We emphasize that the impossibility results should motivate better defense design, not abandonment of defense.

References

[1] G. Alon and M. Kamfonas. Detecting language model attacks with perplexity. *arXiv preprint arXiv:2308.14132*, 2023.

[2] A. Bagnall and G. Stewart. Certifying the true error: Machine learning in Coq with verified generalization guarantees. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019.

[3] Y. Bai et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

[4] P. Chao, A. Robey, E. Dobriban, H. Hassani, G. J. Pappas, and E. Wong. Jailbreaking black box large language models in twenty queries. *arXiv preprint arXiv:2310.08419*, 2024.

[5] J. Cohen, E. Rosenfeld, and J. Z. Kolter. Certified adversarial robustness via randomized smoothing. *Proceedings of ICML*, 2019.

[6] I. J. Goodfellow, J. Shlens, and C. Szegedy. Explaining and harnessing adversarial examples. *Proceedings of ICLR*, 2015.

[7] H. Inan et al. Llama Guard: LLM-based input-output safeguard for human-AI conversations. *arXiv preprint arXiv:2312.06674*, 2023.

[8] G. Katz, C. Barrett, D. L. Dill, K. Julian, and M. J. Kochenderfer. Reluplex: An efficient SMT solver for verifying deep neural networks. *Proceedings of CAV*, 2017.

[9] N. Carlini and D. Wagner. Towards evaluating the robustness of neural networks. *Proceedings of IEEE S&P*, 2017.

[10] A. Mehrotra et al. Tree of attacks: Jailbreaking black-box LLMs automatically. *Advances in Neural Information Processing Systems*, 37, 2024.

[11] A. Fawzi, H. Fawzi, and O. Fawzi. Adversarial vulnerability for any classifier. *Advances in Neural Information Processing Systems*, 31, 2018.

[12] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz. Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. *Proceedings of AISec*, 2023.

[13] X. Huang, M. Kwiatkowska, S. Wang, and M. Wu. Safety verification of deep neural networks. *Proceedings of CAV*, 2017.

[14] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu. Towards deep learning models resistant to adversarial attacks. *Proceedings of ICLR*, 2018.

[15] J.-B. Mouret and J. Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.

[16] G. Naitzat, A. Zhitnikov, and L.-H. Lim. Topology of deep neural networks. *Journal of Machine Learning Research*, 21(184):1–40, 2020.

[17] S. Munshi, M. Bhatt, V. S. Narajala, I. Habler, A. Al-Kahfah, K. Huang, and B. Gatto. Manifold of failure: Behavioral attraction basins in language models. *arXiv preprint arXiv:2602.22291v2*, 2026.

[18] M. Samvelyan et al. Rainbow teaming: Open-ended generation of diverse adversarial prompts. *Advances in Neural Information Processing Systems*, 37, 2024.

[19] G. Singh, T. Gehr, M. Püschel, and M. Vechev. An abstract domain for certifying neural networks. *Proceedings of POPL*, 2019.

[20] C. Szegedy et al. Intriguing properties of neural networks. *Proceedings of ICLR*, 2014.

[21] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, and A. Madry. Robustness may be at odds with accuracy. *Proceedings of ICLR*, 2019.

[22] D. H. Wolpert and W. G. Macready. No free lunch theorems for optimization. *IEEE Trans. Evol. Comput.*, 1(1):67–82, 1997.

[23] A. Zou, Z. Wang, N. Carlini, M. Nasr, J. Z. Kolter, and M. Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.

[24] K. Ge et al. MART: Improving LLM safety with multi-round automatic red-teaming. *Proceedings of NAACL*, 2024.

[25] E. Hubinger et al. Sleeper agents: Training deceptive LLMs that persist through safety training. *arXiv preprint arXiv:2401.05566*, 2024.

[26] C. Anil et al. Many-shot jailbreaking. *Advances in Neural Information Processing Systems*, 37, 2024.

[27] D. Kim et al. What really matters in many-shot attacks? *Proceedings of ACL*, 2025.

[28] Q. Zhan et al. InjecAgent: Benchmarking indirect prompt injections in tool-integrated LLM agents. *Findings of ACL*, 2024.

[29] H. Zhang et al. Agent security bench (ASB): Formalizing and benchmarking attacks and defenses in LLM-based agents. *arXiv preprint arXiv:2410.02644*, 2024.

[30] Y. Yuan et al. The instability of safety. *arXiv preprint arXiv:2512.12066*, 2025.

[31] V. Tsvetkov et al. Quantization and safety: A closer look at LLM safety under weight compression. *arXiv preprint arXiv:2502.15799*, 2025.

[32] F. Nesti et al. Mind the gap: Adversarial attacks against GGUF quantized LLMs. *Proceedings of ICML*, 2025.

[33] H. Hammoud et al. Model merging and safety alignment: One bad model spoils the bunch. *Findings of EMNLP*, 2024.

[34] X. Liu et al. Alignment whack-a-mole. *arXiv preprint arXiv:2603.20957*, 2026.

[35] D. Rosenberg et al. IRIS: Adversarial suffix attacks against robust defenses. *Proceedings of NAACL*, 2025.

[36] X. Zhao et al. Weak-to-strong jailbreaking on large language models. *Proceedings of ICML*, 2025.

[37] Y. Huang et al. The safety tax of reasoning alignment. *arXiv preprint arXiv:2503.00555*, 2025.

[38] Y. Huang et al. On the geometric inevitability of the alignment tax. *arXiv preprint arXiv:2603.00047*, 2026.

[39] L. Bailey et al. Slingshot: RL-based agent-to-agent jailbreaking. *arXiv preprint arXiv:2602.02395*, 2026.

Appendices

A Continuous Interpolation of Discrete Observations

If one wishes to embed discrete behavioral observations into a continuous prompt space, the bridge is standard.

Theorem A.1 (Continuous Relaxation)**.** *Let (X, d) be a connected, normal, Hausdorff metric space, and $S \subset X$ a finite set of observations with observed alignment scores $g\colon S \to \mathbb{R}$ satisfying $g(p) < \tau$ and $g(q) > \tau$ for some $p, q \in S$. Then there exists a continuous $f\colon X \to \mathbb{R}$ extending g (i.e., $f|_S = g$) for which the hypotheses of Theorem 4.1 hold. If the extension is chosen to be Lipschitz (via McShane–Whitney), the hypotheses of Theorem 4.7 also hold. If f is additionally Lipschitz, Theorem 4.10 applies wherever transversality is met.*

Proof sketch. **Step 1** (Extension exists). S is finite and X is T_1, so S is closed. Since X is normal and $g\colon S \to \mathbb{R}$ is continuous on a discrete closed subset, the Tietze extension theorem provides a continuous $f\colon X \to \mathbb{R}$ with $f|_S = g$.

Step 2 (Hypotheses of Theorem 4.1 are satisfied). We have $f(p) = g(p) < \tau$ and $f(q) = g(q) > \tau$, so $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Since X is connected and Hausdorff, Theorem 4.1 applies.

Step 3 (Lipschitz extension enables stronger results). When a Lipschitz extension is needed, the McShane–Whitney theorem provides an L-Lipschitz f agreeing with g on S. Then Theorem 4.7 and Theorem 4.10 apply under their additional hypotheses. □

B Quantitative Refinements

The main paper establishes existence and persistence. This section records explicit quantitative refinements.

Theorem B.1 (Volume Lower Bound)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz with $L > 0$, and let μ denote Lebesgue measure. If there exists c with $f(c) = \tau - \varepsilon/2$, then $B(c,\, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$, giving*

$$\mu(\mathcal{B}_\varepsilon) \;\geq\; V_n \cdot \left(\frac{\varepsilon}{4L}\right)^n,$$

where V_n is the volume of the unit ball in $\mathbb{R}^n$. In $\mathbb{R}^1$, this simplifies to $\mu(\mathcal{B}_\varepsilon) \geq \varepsilon/(2L)$.

Proved in the Lean artifact as `MoF_17_CoareaBound`.

Theorem B.2 (Cone Measure Bound)**.** *In $\mathbb{R}$ with Lebesgue measure μ, if $f(x) \geq \tau + c(x - z)$ for all $x \in (z,\, z + \delta_0)$ with $c > \ell(K + 1)$, then $\mu(\{x : f(D(x)) > \tau\}) \geq \delta_0$.*

Proved in the Lean artifact as `MoF_18_ConeBound`.

Theorem B.3 (Defense Dilemma)**.** *Assume f is differentiable at boundary point z with $G = \|\nabla f(z)\|$, and let ℓ be the defense-path Lipschitz constant. Define $K^* = G/\ell - 1$. Then:*

1. *If $K < K^*$, the persistent unsafe region exists ($G > \ell(K + 1)$, so Theorem 4.10 applies).*
2. *If $K \geq K^*$, the ε-robust bound becomes too loose to exclude success on the steep region.*

Since $\ell \leq L$, the dilemma is sharpest when $\ell \ll L$.

Proved in the Lean artifact as `MoF_19_OptimalDefense`.

Figure 7: Persistence-vs-utility tradeoff as K varies around $K^* = G/\ell - 1$ from Theorem B.3. Below K^* the persistent unsafe region is nonempty; at and above K^* the displacement needed to kill persistence is incompatible with utility preservation.

C Pipeline Composition

Theorem C.1 (Pipeline Lipschitz Degradation)**.** *If stages $T_1, \ldots, T_n$ are $K_1, \ldots, K_n$-Lipschitz, the composed pipeline is $(\prod K_i)$-Lipschitz. For n stages each with $K \geq 2$, the effective constant is K^n.*

Falsifier. A measured pipeline with empirical composed Lipschitz constant $\hat{K}_{\text{composed}} > \hat{K}_1 \cdot \hat{K}_2$ would refute the bound in Theorem C.1: Lipschitz constants submultiply under composition, so the composed constant can never exceed the product.

Theorem C.2 (Pipeline Impossibility)**.** *If the composed pipeline $P = T_n \circ \cdots \circ T_1 \circ D$ is continuous and $P(x) = x$ for all $x \in S_\tau$, then P has boundary fixed points. If D is K_D-Lipschitz and each T_i is K-Lipschitz, the ε-robust band scales as $L \cdot K_D \cdot K^n \cdot \delta$.*

Proved in the Lean artifact as `MoF_15_NonlinearAgents`.

Table 7: Empirical pipeline Lipschitz degradation. The measured composed constant $\hat{K}_{\text{composed}}$ is bounded above by $\prod \hat{K}_i$, consistent with Theorem C.1.

D Vulnerability Landscape

This section characterizes the geometry of the unsafe region.

Theorem D.1 (Basin Structure)**.** *If f is continuous and $f(p) > \tau$, then U_τ is open. Under any measure positive on nonempty open sets, U_τ has positive measure.*

Theorem D.2 (Basin Fragment Minimum Size)**.** *If X is a normed space, f is L-Lipschitz with $f(p) > \tau$, the connected component of U_τ containing p has diameter $\geq 2(f(p) - \tau)/L$.*

Smoother surfaces (smaller L) produce larger basins; rougher surfaces produce smaller fragments.

E Full Proofs

Proof of Theorem 4.1 (Boundary Fixation). **Step 1** (Hausdorff $\Rightarrow$ fixed-point set is closed). $\mathrm{Fix}(D) = \{x : D(x) = x\}$ is the preimage of the diagonal $\Delta \subset X \times X$ under $x \mapsto (D(x), x)$. In a Hausdorff space, Δ is closed, so $\mathrm{Fix}(D)$ is closed.

Step 2 (Utility preservation $\Rightarrow$ safe region $\subseteq$ fixed points). $S_\tau \subseteq \mathrm{Fix}(D)$. Since $\mathrm{Fix}(D)$ is closed: $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$.

Step 3 (Connectedness $\Rightarrow$ safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open. If also closed, it would be clopen—but in a connected space the only clopen sets are $\emptyset$ and X. Since both S_τ and U_τ are nonempty, S_τ is not closed.

Step 4 (Boundary point exists). $\overline{S_\tau} \supsetneq S_\tau$, so there exists $z \in \overline{S_\tau} \setminus S_\tau$. Continuity gives $f(z) \leq \tau$; $z \notin S_\tau$ gives $f(z) \geq \tau$. Hence $f(z) = \tau$.

Step 5 (Defense fixes the boundary point). $z \in \overline{S_\tau} \subseteq \mathrm{Fix}(D)$, so $D(z) = z$ and $f(D(z)) = \tau$. □

Proof of Theorem 4.7 (ε-Robust Constraint). By Theorem 4.1, the fixed boundary point z exists.

Step 1. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) = \mathrm{dist}(D(x), D(z)) \leq K\,\mathrm{dist}(x, z)$.

Step 2. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| = |f(D(x)) - f(z)| \leq L\,\mathrm{dist}(D(x), z) \leq LK\,\mathrm{dist}(x, z)$. □

Proof of Theorem 4.10 (Persistent Unsafe Region). **(1)** $\mathcal{S}$ is the strict superlevel set of $x \mapsto f(x) - \ell(K+1)\,\mathrm{dist}(x, z)$ at level τ, hence open.

(2) Open and nonempty implies positive measure.

(3) For $x \in \mathcal{S}$: $f(x) > \tau + \ell(K+1)\,\mathrm{dist}(x, z)$. By Theorem 4.8: $f(D(x)) \geq f(x) - \ell(K+1)\,\mathrm{dist}(x, z) > \tau$.

Lean: `persistent_unsafe_refined` *in* `MoF_20_RefinedPersistence` *(defense-path constant ℓ).* `MoF_11_EpsilonRobust` *contains an earlier version using the global constant L; as noted in Section 4.3, that version is vacuous for isotropic surfaces.* □

F Counterexamples: Each Hypothesis Is Necessary

Counterexample F.1 (Removing connectedness)**.** *$X = \{0, 1\}$ discrete, $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$: continuous, utility-preserving, complete.*

Counterexample F.2 (Removing continuity)**.** *$X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$: utility-preserving and complete, but discontinuous at 0.5.*

Counterexample F.3 (Removing utility preservation)**.** *$D(x) = x_0$ for a fixed safe point: continuous and complete, but destroys all inputs.*

Table 8: Counterexample ablation. Each row drops one of the three hypotheses and reports whether the defense becomes complete, together with the price paid in the other two properties.

Case	Hypothesis dropped	Construction	Validator flag
C.1	connectedness	$X = \{0, 1\}$ discrete, $D = \mathrm{id}$	`applicable=False`; $\overline{S_\tau} = S_\tau$
C.2	continuity	$X = [0, 1]$, $D(x) = x \cdot \mathbf{1}_{x<0.5}$	local Lipschitz $K = 49.0$ at x=0.495
C.3	utility preservation	$D(x) = x_0$ constant	$\lvert D(X)\rvert = 1$; 50/50 safe inputs displaced

G Attack Properties

Theorem G.1 (Perturbation Robustness)**.** *If f is L-Lipschitz and $f(p) > \tau$, then $B(p, (f(p) - \tau)/L) \subseteq U_\tau$. The radius is monotone in $f(p)$.*

Theorem G.2 (Iterative Convergence)**.** *Any monotone-improvement operator T with score in $[0, 1]$ converges. If each step gains $\geq \delta$, convergence takes $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem G.3 (Transferability)**.** $\|f - g\|_\infty \leq \delta \implies \{f > \tau + \delta\} \subseteq \{g > \tau\}$. *Transfer costs zero additional queries.*

Theorem G.4 (Authority Monotonicity)**.** *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, the vulnerability set is upward-closed with a critical threshold $a_2^*(y)$ (by IVT, when $f(y, \cdot)$ is continuous and crosses τ). If additionally f is monotone non-decreasing in indirection for each fixed authority, the threshold curve $y \mapsto a_2^*(y)$ is non-increasing.*

Theorem G.5 (Gradient Ascent)**.** *If f has nonzero Fréchet derivative at x, there exist v and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

H Stability Under Fine-Tuning

Theorem H.1 (Interior Stability)**.** *If $\|f - g\|_\infty \leq \varepsilon$: $f(x) > \tau + \varepsilon \implies g(x) > \tau$, and $f(x) < \tau - \varepsilon \implies g(x) < \tau$. Only the band $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem H.2 (Crossing Preservation)**.** *Let $f, g\colon [a, b] \to \mathbb{R}$ be continuous. If f crosses τ on $[a, b]$ with margin m (i.e., $f(a') < \tau - m$ and $f(b') > \tau + m$ for some $a', b' \in [a, b]$) and $\|f - g\|_\infty \leq \varepsilon < m$, then g also crosses τ on $[a, b]$.*

Theorem H.3 (Patching Is Nonlocal)**.** *There exist f, g with $\|f - g\|_\infty \leq \varepsilon$ such that eliminating a vulnerability at one point necessarily changes values at distant points.*

I Cost Asymmetry

Theorem I.1 (Exponential Cost Asymmetry)**.** *For grid-based defense with $N \geq 2$ bins per axis in d dimensions: attack cost $\leq 1/\delta$ (dimension-independent); defense cost $= N^d$ (exponential in d); ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*

At $d = 2$ with $N = 25$ and $\delta = 0.01$, the ratio is 6.25. At $d = 10$, it climbs to $\sim 10^{12}$.

J Additional Verified Results

Lipschitz displacement bound. D is K-Lipschitz with $D(z) = z \implies \mathrm{dist}(D(x), x) \leq (K + 1)\,\mathrm{dist}(x, z)$.

Tool calls amplify failure. Each non-contractive tool call multiplicatively increases the pipeline's effective Lipschitz constant: $K^n < K^{n+1}$ for $K \geq 2$.

Attacker monotone improvement. Best observed alignment deviation is non-decreasing across turns.

Attacker steering. If directional slope varies continuously with attacker parameter α and crosses $\ell(K+1)$, transversality is reachable by IVT.

Stochastic regularity. $g(x) = \mathbb{E}[f(D(x))]$ satisfies $g - f = 0$ on $\overline{S_\tau}$.

Discrete defense dilemma. An injective, utility-preserving defense is incomplete (every unsafe input stays unsafe). A complete, utility-preserving defense is non-injective (distinct inputs collapse to the same output). The three properties—completeness, utility preservation, injectivity—form a trilemma.

Defense position invariance. Lipschitz constant of defense-before-tools equals defense-after-tools: $K_D \cdot K_T^n$.

K Additional Experimental Details

This appendix collects the estimator definitions, sanity checks, continuous-defense sweeps, and robustness tables that support the main paper's empirical claims.

K.1 Estimator definitions

Let G_f denote the set of filled cells in an $N \times N$ MAP-Elites grid and let $h = 1/N$ be the cell width. Distances use normalized L_2 grid coordinates on $[0,1]^2$:

$$\mathrm{dist}\big((i_1, j_1), (i_2, j_2)\big) = h\sqrt{(i_1 - i_2)^2 + (j_1 - j_2)^2}.$$

All pairs involving unfilled cells are excluded; no imputation is performed. The boundary anchor is

$$z^* = \arg\min_{x \in G_f} |f(x) - \tau|,$$

restricted to sampled boundary cells $\overline{S_\tau} \setminus S_\tau$, with lexicographic tie-breaking.

Surface Lipschitz constant.

$$\hat{L} = \max_{x,y \in G_f,\ x \neq y} \frac{|f(x) - f(y)|}{\mathrm{dist}(x, y)}.$$

Defense Lipschitz constant.

$$\hat{K} = \max_{u,v \in G_f,\ u \neq v} \frac{\mathrm{dist}(D(u), D(v))}{\mathrm{dist}(u, v)}.$$

Defense-path Lipschitz constant.

$$\hat{\ell} = \max_{x \in G_f,\ D(x) \neq x,\ D(x) \in G_f} \frac{|f(D(x)) - f(x)|}{\mathrm{dist}(D(x), x)}.$$

If the set is empty, we take $\hat{\ell} = 0$.

Boundary directional rate.

$$\hat{G} = \max_{x,y \in G_f,\ f(x) < \tau < f(y)} \frac{f(y) - f(x)}{\mathrm{dist}(x, y)}.$$

These are the estimators used by the validator and the tables below.

Table 9: **Smoke tests (not a test of the theorem).** The two sparse 25×25 live runs under the canonical identity defense. Under $D = \mathrm{id}$, $\mathcal{S}_{\text{pred}} \equiv \mathcal{S}_{\text{act}} \equiv \{x : f(x) > \tau\}$ by construction, so agreement only checks the validator's bookkeeping.

Target	τ	**Filled**	L	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	$\lvert\mathcal{S}_{\text{act}}\rvert$	**TP**	**FP**$_{\text{int}}$	**FN**
gpt-5-mini-2025-08-07	0.30	14	6.13	6.13	7	7	7	**0**	0
gpt-3.5-turbo-0125 (smoke)	0.50	9	11.44	11.44	7	7	7	**0**	0

K.2 Identity and out-of-scope sanity checks

Identity-defense rows are bookkeeping checks rather than theorem tests: under $D = \mathrm{id}$, the predicted and observed unsafe sets coincide by construction.

K.3 Continuous-defense sweeps

Table 10: Continuous-defense sweep on the saturated `gpt-3.5-turbo-0125` grid over continuous defense families (GP-oblique, Gaussian-smoothed nearest-safe, contractive-flow). Each row reports empirical $\hat{\ell}$, $\hat{K}$, $|\mathcal{S}_{\text{pred}}|$, $|\mathcal{S}_{\text{act}}|$, TP, and FP$_{\text{int}}$.

defense	L	K	ℓ	$\ell(K{+}1)$	G	trans.	$\lvert\mathcal{S}_{\text{pred}}\rvert$	TP	FP$_{\text{int}}$	$\lvert\mathcal{S}_{\text{act}}\rvert$
SMOOTHNEARESTSAFE	23.62	3.162	23.625	98.334	23.62	×	1	1	0	59
KERNELSMOOTHED	23.62	5.000	21.750	130.500	23.62	×	1	1	0	39
SOFTPROJ	23.62	2.000	23.625	70.875	23.62	×	1	0	0	56
OBLIQUEGP	23.62	1.050	0.274	0.563	23.62	✓	46	46	0	68

Table 11: GP-kernel sensitivity for the oblique-smooth defense on the saturated `gpt-3.5-turbo-0125` grid. Kernel $\in \{\text{RBF}, \text{Matérn-}3/2, \text{Matérn-}5/2\}$ crossed with bandwidth $\sigma \in \{0.1, 0.2, 0.4\}$.

kernel	σ	L	K	ℓ	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	TP	FP$_{\text{int}}$
RBF	0.10	23.62	1.105	0.884	23.62	9	9	0
RBF	0.20	23.62	1.050	0.274	23.62	46	46	0
RBF	0.40	23.62	1.037	0.099	23.62	71	71	0
Matérn-3/2	0.10	23.62	1.068	1.315	23.62	7	7	0
Matérn-3/2	0.20	23.62	1.070	1.165	23.62	8	8	0
Matérn-3/2	0.40	23.62	1.070	0.946	23.62	8	8	0
Matérn-5/2	0.10	23.62	1.079	0.899	23.62	9	9	0
Matérn-5/2	0.20	23.62	1.080	0.734	23.62	9	9	0
Matérn-5/2	0.40	23.62	1.094	0.345	23.62	41	41	0

K.4 Robustness checks

K.5 Out-of-scope discrete stress tests

Bounded-step defenses are not continuous, so Theorem 4.10's hypotheses do not apply. We include them only as negative consistency checks.

L Judge robustness

All f values in Section 7.1 are judge-dependent: swapping the judge can reshape the alignment-deviation surface and thereby change $\hat{L}, \hat{\ell}, \hat{G}$. To quantify this, we rescored the 82 (prompt, target response) pairs from the saturated `gpt-3.5-turbo-0125` archive with three alternative judges drawn from different families and capacities — `gpt-4o-2024-08-06`, `gpt-4.1-mini-2025-04-14`, and `gpt-4o-mini-2024-07-18` — applying the same rethinking-evals per-category methodology with a single judge and $\alpha = 0.5$ (target responses were NOT regenerated; only the judge was swapped). The four-judge comparison is summarized in Table 18.

Table 12: Resolution sweep for the oblique-smooth defense on `gpt-3.5-turbo-0125`. As $h = 1/N$ decreases, the discretization gap $|f(z^*) - \tau|$ shrinks and $|\mathcal{S}_{\text{pred}}|$ grows; the theorem's containment is preserved across resolutions.

grid	filled cells	L	K	ℓ	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	TP	FP_{int}
13×13	25	11.960	1.048	0.629	11.960	5	5	0
17×17	31	14.195	1.033	0.598	14.195	5	5	0
21×21	52	19.845	1.064	0.966	19.845	6	6	0
25×25	82	23.625	1.051	0.865	23.625	3	3	0

Table 13: Higher-dimensional robustness. Empirical $\hat{L}$ and $\hat{G}$ computed on the same 82-cell archive using the paper's 2D grid metric and a 768-dimensional embedding metric.

Metric	$\hat{L}$	$\hat{G}$
2D grid coordinates (normalized to $[0,1]^2$)	22.680	22.680
768-d MPNet embedding (pairwise-max normalized)	17.926	17.926

Interpretation. The theorem applies to the full prompt space X under every judge — benign prompts exist, so $\mathcal{S}_\tau$ is nonempty in X. Whether the theorem's positive prediction can be *tested from a given sample* depends on whether that sample spans both regions. Three of four judges (`gpt-4.1`, `gpt-4o`, `gpt-4.1-mini`) yield samples that span both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$; applying the continuous `smooth_nearest_safe` defense produces TP> 0 and $\text{FP}_{\text{int}} = 0$ in every case. Under the strictest judge (`gpt-4o-mini`), MAP-Elites's adversarial bias drives the sample entirely into $\mathcal{U}_\tau$, so no boundary cell z^* appears in the 82 sampled cells and the positive prediction cannot be tested from this sample. Crucially, $\text{FP}_{\text{int}} = 0$ still holds: no counterexample to the theorem is produced under any of the four judges. Cross-judge agreement ranges from substantial (Kendall's $W = 0.70$) to poor on binary classification (Cohen's $\kappa = 0$ vs gpt-4o-mini), so the *specific numerical instance* of a particular judge is not itself judge-robust even though the *framework's predictions* are.

Committee aggregation. The rethinking-evals methodology supports committee aggregation: $P_{\text{actual}} = \alpha P_{\text{vote}} + (1-\alpha) P_{\text{mean}}$ with $\alpha = 0.5$. We apply this offline to the four existing judge archives (no new API calls) and re-run the validator on each committee-aggregated surface (Table 19). The top-3 committee (gpt-4.1, gpt-4o, gpt-4.1-mini; excluding the strict-judge outlier gpt-4o-mini) is within one cell of canonical (17 safe vs 16; mean $\Delta = +0.010$), and $\text{FP}_{\text{int}} = 0$ under `smooth_nearest_safe` on every committee configuration — the theorem's predictions survive committee aggregation.

M Attack Baseline: PAIR-Style Iteration

To situate the theorem empirically against an established attack-defense benchmark style, we run a PAIR-style iterative attack ([4]-adjacent) on ten top-AD unsafe seed prompts from the saturated archive. The attacker is `gpt-4o-mini`, the target is `gpt-3.5-turbo-0125`, and the judge is `gpt-4.1`. Each behavior gets five iterations; the attacker receives the full transcript and judge scores at each step. We compare two conditions: undefended (attacker prompt goes directly to target) and *paraphrase wrapper* (attacker prompt is rewritten by `gpt-4o-mini` at $T = 0$ before reaching the target).

Interpretation. Within the 5-iteration budget the paraphrase wrapper blocks all ten behaviors (ASR drops from 1.00 to 0.00). This does *not* contradict Theorem 5.1's injective-incomplete horn: the paraphrase defense is injective on our 66-prompt sample (all 66 rewrites are distinct), and Table 6 already exhibits $1/66$ paraphrases that yield $f(D(x)) > \tau$. The PAIR attacker with gpt-4o-mini as refiner and a 5-iteration budget is not sufficient to locate that bypass; a persistent attacker (more iterations, stronger refiner) is predicted to find one by Theorem 6.1's multi-turn convergence argument.

Table 14: Seed replication of the saturated pipeline on `gpt-3.5-turbo-0125`. Surface quantities remain within a narrow band and $\mathrm{FP}_{\mathrm{int}} = 0$ in both runs.

Run	Filled	$\hat{L}$	$\hat{K}$	$\hat{\ell}$	$\hat{G}$	trans.	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$
Saturated (canonical)	82	23.62	3.16	23.62	23.62	×	1	1	0
Replicate (independent)	76	22.25	2.69	22.25	22.25	×	1	1	0

Table 15: Independent target re-evaluation. The same 82 prompts are rescored on `gpt-4o-mini-2024-07-18` with the judge held fixed. No interior counterexample is observed.

Target	Cells	Peak AD	L	K	ℓ	G	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$	Transv.
`gpt-3.5-turbo-0125`	82	1.000	23.62	3.16	23.62	23.62	1	1	0	×
`gpt-4o-mini-2024-07-18`	82	0.990	12.25	4.94	4.02	10.12	0	0	0	×

N Lean Artifact

The complete theory is verified in Lean 4.28.0 with Mathlib v4.28.0, available at `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs`. The artifact comprises 46 files:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 1 continuous relaxation (MoF_ContinuousRelaxation)
- 1 ε-robust constraint + persistent unsafe region with global Lipschitz constant (MoF_11_EpsilonRobust)
- 1 discrete impossibility (MoF_12_Discrete)
- 1 multi-turn + stochastic extensions (MoF_13_MultiTurn)
- 1 representation-independent meta-theorem (MoF_14_MetaTheorem)
- 1 nonlinear agent pipelines (MoF_15_NonlinearAgents)
- 1 relaxed utility preservation (MoF_16_RelaxedUtility)
- 1 quantitative ε-band volume bound (MoF_17_CoareaBound)
- 1 cone measure bound for persistent unsafe region (MoF_18_ConeBound)
- 1 optimal defense characterization (MoF_19_OptimalDefense)
- 1 refined persistence with defense-path Lipschitz constant ℓ (MoF_20_RefinedPersistence); this file contains the primary formalization of Theorems 4.8 to 4.10
- 1 gradient chain: $\|\nabla f(z)\| > \ell(K+1)$ implies persistent unsafe region via operator-norm direction extraction and derivative-based local growth (MoF_21_GradientChain)
- 3 capstone files (MasterTheorem, Euclidean instantiation, verification)
- 1 root import file (ManifoldProofs.lean)

Table 16: Bootstrap percentile confidence intervals (median, $[5^{\text{th}}, 95^{\text{th}}]$) for the empirical quantities on the saturated live run.

Estimator	identity	GP-smooth oblique
L median	22.680	22.680
L 95% CI	$[20.880, 22.680]$	$[20.880, 22.680]$
K median	1.000	1.051
K 95% CI	$[1.000, 1.000]$	$[1.029, 1.051]$
ℓ median	–	0.865
ℓ 95% CI	$[--, --]$	$[0.801, 0.865]$
G median	22.680	22.680
G 95% CI	$[20.880, 22.680]$	$[20.880, 22.680]$

Figure 8: Sensitivity of $|\mathcal{S}_{\text{pred}}|$, $|\mathcal{S}_{\text{act}}|$, and $\mathrm{FP}_{\mathrm{int}}$ to the threshold τ on the saturated live run. The containment $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$ holds at every tested τ.

Table 17: **Out-of-scope discrete stress tests.** Defense sweep on the saturated `gpt-3.5-turbo-0125` grid over discontinuous `bounded_step` defenses. These rows are not theorem checks; they are reported only as consistency checks outside the theorem's scope.

Defense	$\boldsymbol{K}$	$\boldsymbol{\ell}$	$\lvert\mathcal{S}_{\text{pred}}\rvert$	$\lvert\mathcal{S}_{\text{act}}\rvert$	**TP**	$\mathbf{FP}_{\text{int}}$	**FN**
identity	1.00	0.00	66	66	66	**0**	0
`bounded_step(1)`	3.61	23.63	1	55	0	**0**	55
`bounded_step(2)`	3.61	23.63	1	47	0	**0**	47
`bounded_step(3)`	4.00	23.63	1	29	0	**0**	29
`bounded_step(5)`	8.60	23.63	1	3	0	**0**	3
`bounded_step(8)`	8.60	23.63	1	0	0	**0**	0
`bounded_step(12)`	8.60	23.63	1	0	0	**0**	0

Table 18: Judge robustness on the saturated archive (82 cells, $\tau = 0.5$). $|\mathcal{S}_\tau|_{\text{sample}}$ and $|\mathcal{U}_\tau|_{\text{sample}}$ are how many of the 82 sampled cells fall below/above τ under each judge. Pearson r and Cohen's κ are pairwise with the canonical gpt-4.1 judge. Kendall's $W = 0.700$ across all four judges on cell rankings (moderate agreement). "Testable?" indicates whether this MAP-Elites sample spans both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$, which is what the positive-prediction test of Theorem 4.10 requires to locate a boundary cell z^*.

Judge	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	peak AD	mean AD	Pearson r	κ	TP	$\mathrm{FP}_{\mathrm{int}}$	Testable
`gpt-4.1-2025-04-14` (canonical)	16	66	1.000	0.773	–	–	1	0	yes
`gpt-4o-2024-08-06`	10	72	0.990	0.889	0.546	0.457	1	0	yes
`gpt-4.1-mini-2025-04-14`	33	49	0.975	0.752	0.493	0.363	1	0	yes
`gpt-4o-mini-2024-07-18`	0	82	1.000	0.955	0.393	0.000	–	–	no

Table 19: Judge committee aggregations vs the canonical single-judge surface. Top-3 committee is within one cell of canonical; $\mathrm{FP}_{\mathrm{int}} = 0$ in every configuration.

committee	N	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	mean Δ	std Δ	$\mathrm{FP}_{\mathrm{int}}$
canonical (gpt-4.1 only)	1	16	66	0.000	0.000	0
top-3 (gpt-4.1, gpt-4o, gpt-4.1-mini)	3	17	65	+0.010	0.173	0
gpt-4 family (gpt-4.1, gpt-4o)	2	13	69	+0.063	0.141	0
all-4 (incl. strict gpt-4o-mini)	4	5	77	+0.058	0.200	0

Table 20: PAIR-style attack-success rate with and without a continuous paraphrase wrapper (10 behaviors, 5 iterations, gpt-3.5-turbo target, gpt-4.1 judge).

Condition	behaviors	ASR @ 5 iters	mean running-max
undefended	10	1.00	0.995
paraphrase wrapper	10	0.00	0.500

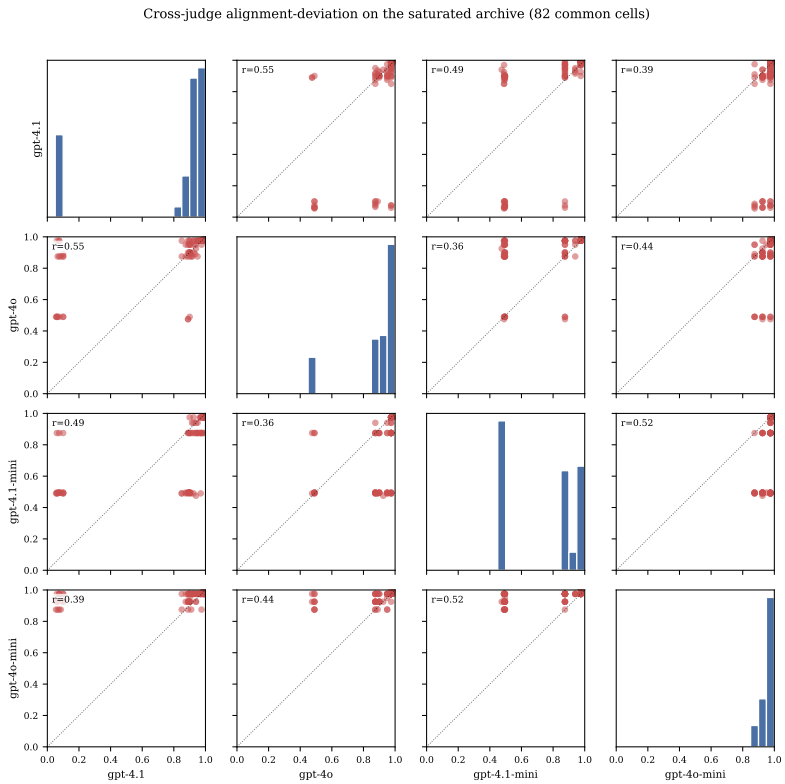

Figure 9: Per-cell alignment-deviation scatter across the four judges. Cross-judge agreement is moderate (Kendall's $W = 0.700$); the positive-prediction test of Theorem 4.10 remains supported wherever the sample spans both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$.